

Enrique Gorosave Meza

AI Systems Engineer | Environmental Intelligence Infrastructure Specialist

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Professional Profile

Systems-oriented engineer with specialized expertise in **Bioengineering** and the design of **applied AI systems for environmental governance**. Founder of **Esoteria Intelligence Infrastructure**, where I architect and deploy platforms that transform unstructured data into actionable, auditable geospatial intelligence. Expert in building automated data pipelines that integrate remote sensing, citizen science, and agentic AI to address complex ecological challenges in marine and terrestrial ecosystems.

Core Research & Technical Skills

- **AI & Machine Learning:** Development of **agentic AI pipelines** using Google Gemini and **VLM (Vision-Language Model)** architectures for environmental auditing.
- **Geospatial Data Engineering:** Advanced use of **GFW API v3**, **BigQuery**, and **gfw-api-python-client** for tracking vessel activity and fishing effort.
- **Spatial Indexing:** Implementation of Uber's **H3 hexagonal grid** (resolutions 5 and 7) for multidimensional risk scoring and environmental modeling.
- **Full-Stack Development:** Expert in **Python**, **FastAPI**, **SQL**, and **Next.js** for deploying interactive research dashboards.
- **Environmental Governance:** Deep knowledge of Mexico's SEMARNAT regulatory framework and Marine Protected Areas (MPA) in the Gulf of California.

Key Conservation Technology Projects

Project Lead | Guardians of the Gulf (Ocean Proto)

2025 – Present

- Developed the **Ocean Proto** platform, a functional geospatial risk analytics prototype that ingests **GFW fishing effort**, encounters, and loitering data.
- Integrated **OBIS marine megafauna occurrence records** for 42 cetacean species with NOAA oceanographic variables into a unified H3 scoring engine.
- Designed a **Composite Risk Score (CRS)** framework to identify hotspots where high biological productivity overlaps with intense human pressure.

Architect | Zohar Agent (Esoteria)

- Engineered an agentic AI pipeline for the automated extraction and structuring of **Mexico's SEMARNAT environmental impact assessments** (Gaceta Ecológica).
- Transformed unstructured government PDFs into auditable geospatial intelligence using **sovereign edge-computing**.

Developer | Denuncia Popular Platform

- Created a citizen science platform for geolocated environmental crime reporting, integrating **Google Gemini AI** to generate legally-certified complaints.

Professional Experience

Systems & Infrastructure Lead | New Fortress Energy (NFE)

March 2021 – October 2024

- Coordinated technical commissioning and systems architecture for critical energy infrastructure in Baja California Sur, ensuring high-availability operations and rigorous data integrity.

Education

- **B.S. in Bioengineering** | Universidad Autónoma de Baja California (UABC).

Publications & Deliverables

- **Composite Risk Score (CRS) Methodology:** Reproducible Python package for multi-criteria anthropogenic pressure modeling.
- **Interactive Risk Dashboards:** Open-source platforms for real-time visualization of marine megafauna-fleet interactions.